

Sarah K. Hu, Ph.D.
Assistant Professor
Texas A&M University
Department of Oceanography
1204 Eller O&M Building, TAMU 3146

skhu@tamu.edu
858-945-5274
she/her
ORCID: 0000-0002-4439-1360
Lab website: <https://shu251.github.io/sarah-hu/>

PROFESSIONAL PREPARATION

2023 - present	Assistant Professor Texas A&M University, College Station, TX <i>Department of Oceanography</i>
2020 - 2022	Postdoctoral Research Investigator Woods Hole Oceanographic Institution, Woods Hole, MA <i>Marine Chemistry & Geochemistry</i> , PI: Dr. Julie Huber
2018 - 2020	Postdoctoral Research Fellow, Center for Dark Energy Biosphere Investigations Woods Hole Oceanographic Institution, Woods Hole, MA <i>Marine Chemistry & Geochemistry</i> , PI: Dr. Julie Huber
2018	Postdoctoral Research Associate University of Southern California, Los Angeles, CA <i>Marine Environmental Biology</i> , PI: Dr. David Caron
2018	Ph.D. Marine Biology and Biological Oceanography University of Southern California, Los Angeles, CA <i>Marine Environmental Biology</i>
2011	B.S. Aquatic & Fishery Sciences, Minor: Microbiology University of Washington, Seattle, WA

PUBLICATIONS

1. Rizzo, R., Scott, N. M., Kubik, B. C., **Hu, S. K.**, Huber, J. A., & Holden, J. F. (2026). *Methanothermococcus jasoni* sp. nov., a novel thermophilic methanogen from subseafloor hydrothermal vent fluids of Axial Seamount, Juan de Fuca Ridge. ISME Journal;76(3) <https://doi.org/10.1099/ijsem.0.007098>.
2. Gleich, S.J., Mesrop, L.Y., Cram, J.A., Weissman, J.L., **Hu, S.K.**, Yeh, Y.-C., et al. (2025) With a little help from my friends: importance of protist-protist interactions in structuring marine protistan communities in the San Pedro Channel. mSystems e0104524.
3. Krinos, A.I., Mars Brisbin, M., **Hu, S.K.**, Cohen N.R., Ryneerson, T.A., Follows, M.J., Schulz, F., & Alexander, H. (2024) Missing Microbial Eukaryotes and Misleading Meta-Omic Conclusions. Nature Communications; 15 (1): 9873. <https://doi.org/10.1038/s41467-024-52212-w>.
4. **Hu S. K.**, Anderson R.E., Pachiadaki M.G., Edgcomb V.P., Serres M.H., Sylva S.P., German, C.R., Seewald, J.S., Lang, S.Q., & Huber, J.A. (2024). Microbial eukaryotic predation pressure and biomass at deep-sea hydrothermal vents. ISME Journal; 18.
5. Gleich S.J., **Hu S.K.**, Krinos A.I., & Caron D.A. (2024) Protistan community composition and metabolism in the North Pacific Subtropical Gyre: Influences of mesoscale eddies and depth. Environmental Microbiology; 26: e16556.
6. Beckett S.J., Demory D., Coenen A.R., Casey J.R., Dugenne M., Follett C.L., Connell P., Carlson M.C.G., **Hu S.K.**, & 12 others. (2024) Disentangling top-down drivers of mortality underlying diel population dynamics of *Prochlorococcus* in the North Pacific Subtropical Gyre. Nature Communications; 15: 2105.
7. Ollison, G.A., **Hu, S.K.**, Hopper, J.V., Stewart, B.P., Beatty, J.L., & Caron, D.A. (2023). Physiology governing diatom vs. dinoflagellate bloom and decline in coastal Santa Monica Bay. Frontiers in Microbiology; 14. <https://doi.org/10.3389/fmicb.2023.1287326>
8. Alexander, H., **Hu, S.K.**, Krinos, A.I., Pachiadaki, M., Tully, B.J., Neely, C.J., & Reiter, T. (2023) Eukaryotic genomes from a global metagenomic data set illuminate trophic modes and biogeography of ocean plankton. mBio; e0167623. DOI: 10.1128/mbio.01676-23

9. **Hu, S. K.**, Smith, A., Anderson, R., Sylva, S., Setzer, M., Steadmon, M., Frank, K., Chan, E., Lim, D., German, C., Breier, J.A., Lang, S.Q., Butterfield, D., Fortunato, C.S., Seewald, J., & Huber, J.A. (2022) Globally-distributed microbial eukaryotes exhibit endemism at deep-sea hydrothermal vents. *Molecular Ecology*; 32(23). DOI: 10.1111/mec.16745
10. Hammond, S.W., Lodolo, L., **Hu, S.K.**, & Pasulka, A.L. (2022) Methodological 'lenses' influence the characterization of phytoplankton dynamics in a coastal upwelling ecosystem. *Environmental Microbiology Reports*; DOI: 10.1111/1758-2229.13116.
11. Cohen, N.R., Alexander, H., Krinos, A.I., **Hu, S.K.**, and Lampe, R.H. (2022) Marine Microeukaryote Metatranscriptomics: Sample Processing and Bioinformatic Workflow Recommendations for Ecological Applications. *Frontiers in Marine Science*; 9: 867007.
12. Ollison, G.A., **Hu, S.K.**, Hopper, J.V., Stewart, B.P., Smith, J., Beatty, J.L., Rink, L.K., & Caron, D.A. (2022) Daily dynamics of contrasting spring algal blooms in Santa Monica Bay (central Southern California Bight). *Environmental Microbiology*; 1462-2920.16137.
13. Tully, B.J., Buongiorno, J., Cohen, A.B., Cram, J.A., Garber, A.I., **Hu, S.K.**, & 10 others. (2021) The Bioinformatics Virtual Coordination Network: an open-source and interactive learning environment. *Frontiers in Education*; 6:711618. DOI: 10.3389/educ.2021.711618
14. **Hu, S.K.**, Herrera, E., Smith, A., Pachiadaki, M.G., Edgcomb, V.P., Sylva, S.P., Chan E.W., Seewald, J.S., German, C.R., & Huber, J.A. (2021) Protistan grazing impacts microbial communities and carbon cycling at deep-sea hydrothermal vents. *Proceedings of the National Academy of Sciences*; 118(29). DOI: 10.1073/pnas.2102674118
15. Ollison G., **Hu, S.K.**, Mesrop, L.Y., Delong, E. & Caron, D. A. (2021) Come Rain or Shine: Depth Not Season Shapes the Protistan Community at Station ALOHA in the North Pacific Subtropical Gyre. *Deep-Sea Research Part I: Oceanographic Research Papers*; 120, 103494. DOI: 10.1016/j.dsr.2021.103494
16. Coesel, S.N., Durham, B.P., Groussman, R.D., **Hu, S.K.**, Caron, D.A., Morales, R.L., Ribalet, F., & Armbrust, E. V. (2021) Diel transcriptional oscillations of light-sensitive regulatory elements in open-ocean eukaryotic plankton communities. *Proceedings of the National Academy of Sciences*; 118(12). DOI: 10.1073/pnas.2011038118
17. Krinos, A., **Hu, S.K.**, Cohen, N.R. & Alexander, H. (2021) EUKulele: Taxonomic annotation of the unsung eukaryotic microbes. *Journal of Open Source Software*; 6(57), 2817. DOI: 10.21105/joss.02817
18. Coenen, A.R., **Hu, S.K.**, Luo, E., Muratore, D., and Weitz, J.S. (2020) A Primer for Microbiome Time-Series Analysis. *Frontiers in Genetics*; 11(310). DOI: 10.3389/fgene.2020.00310
19. Boeuf, D., Edwards, B.R., Eppley, J.M., **Hu, S.K.**, Poff, K.E., Romano, A.E., Caron, D.A., Karl, D.M., & DeLong, E.F. (2019) Biological composition and microbial dynamics of sinking particulate organic matter at abyssal depths in the oligotrophic open ocean. *Proceedings of the National Academy of Sciences*; 116(24): 11824–11832. DOI: 10.1073/pnas.1903080116
20. Liu, Z., Mesrop, L.Y., **Hu, S.K.**, & Caron, D.A. (2019) Transcriptome of *Thalassicolla nucleata* Holobiont Reveals Details of a Radiolarian Symbiotic Relationship. *Frontiers in Marine Science* 6(284). DOI: 10.3389/fmars.2019.00284.
21. Pasulka, A., **Hu, S.K.**, Countway, P.D., Coyne, K.J., Cary, S.C., Heidelberg, K.B., & Caron, D.A. (2019) SSU rRNA Gene Sequencing Survey of Benthic Microbial Eukaryotes from Guaymas Basin Hydrothermal Vent. *Journal of Eukaryotic Microbiology*; 66(4):637-653. DOI: 10.1111/jeu.12711
22. Caron, D.A. & **Hu, S.K.** (2019) Are We Overestimating Protistan Diversity in Nature? *Trends in Microbiology*; 27(3): 197–205. DOI: 10.1016/j.tim.2018.10.009
23. **Hu, S.K.**, Liu, Z., Alexander, H., Campbell, V., Connell, P.E., Dyhrman, S.T., Heidelberg, K.B., & Caron, D.A. (2018) Shifting metabolic priorities among key protistan taxa within and below the euphotic zone. *Environmental Microbiology*; 20: 2865–2879.
24. **Hu, S.K.**, Paige E. Connell, Mesrop, L.Y., & Caron, D.A. (2018) A Hard Day's Night: Diel Shifts in Microbial Eukaryotic Activity in the North Pacific Subtropical Gyre. *Frontiers in Marine Science*; 5(251). DOI: 10.3389/fmars.2018.00351
25. Liu, Z., **Hu, S.K.**, Campbell, V., Tatters, A.O., Heidelberg, K.B., & Caron, D.A. (2017) Single-cell transcriptomics of small microbial eukaryotes: limitations and potential. *ISME Journal*; 11: 1282–1285.

26. Connell, P.E., Campbell, V., Gellene, A.G., **Hu, S.K.**, & Caron, D.A. (2017) Planktonic food web structure at a coastal time-series site: II. Spatiotemporal variability of microbial trophic activities. *Deep-Sea Research Part I: Oceanographic Research Papers*; 121: 210–223.
27. **Hu, S.K.**, Campbell, V., Connell, P., Gellene, A.G., Liu, Z., Terrado, R., & Caron, D.A. (2016) Protistan diversity and activity inferred from RNA and DNA at a coastal ocean site in the eastern North Pacific. *FEMS Microbiology Ecology*; 92(4). DOI: 10.1093/femsec/fiw050
28. **Hu, S.K.**, Liu, Z., Lie, A.A.Y., Countway, P.D., Kim, D.Y., Jones, A.C., & 5 others. (2015) Estimating Protistan Diversity Using High-Throughput Sequencing. *Journal of Eukaryotic Microbiology*; 62: 688–693. DOI: 10.1111/jeu.12217
29. Lie, A.A.Y., Liu, Z., **Hu, S.K.**, Jones, A.C., Kim, D.Y., Countway, P.D., & 6 others. (2014) Investigating Microbial Eukaryotic Diversity from a Global Census: Insights from a Comparison of Pyrotag and Full-Length Sequences of 18S rRNA Genes. *Applied and Environmental Microbiology*; 80(14):4363. DOI: 10.1128/AEM.00057

PRESS & NON-REFEREED PUBLICATIONS

1. *River to Remedy: DOE Program Supports Texas A&M Research on Gulf Coast Microbes*. (2026) News feature announcing Department of Energy award. <https://artsci.tamu.edu/news/2026/03/river-to-remedy-doe-program-supports-texas-am-research-on-gulf-coast-microbes.html>
2. *Live from the Gulf*. (2025) Ship-to-shore live stream event: TAMU Oceanography sponsored event to share at-sea field work from the *RV Point Sur*. <https://www.youtube.com/watch?v=yRR1ScflyCs>
3. *A Microbial Oceanography Professor Making An Impact Beyond The Ocean*. (2025) Profile. TAMU College of Arts & Sciences News. Link: <https://artsci.tamu.edu/news/2025/03/a-microbial-oceanography-professor-making-an-impact-beyond-the-ocean.html>
4. Thomas, T. 2023. Microbial eukaryote diversity at hydrothermal vents. *Nature Reviews Earth & Environment*, 4(4):215–215, <https://doi.org/10.1038/s43017-023-00417-w>. Featured research highlight from Hu et al. 2022 *Molecular Ecology*.
5. Blog posts highlighting our research expedition to Axial Seamount. July 2023. *Mission Accomplished!* <https://ndsf.whoi.edu/mission-accomplished/> *The possibilities are endless* <https://ndsf.whoi.edu/the-possibilities-are-endless/> *Life and Dinner among the microbes* <https://ndsf.whoi.edu/life-and-dinner-among-the-microbes/>
6. *Study Examines the Role of Deep-Sea Microbial Predators at Hydrothermal Vents*. (2021) Press release associated with publication of paper in PNAS. <https://www.whoi.edu/press-room/news-release/study-examines-the-role-of-deep-sea-microbial-predators-at-hydrothermal-vents/>
7. Lim, Darlene S.S., Raineault, N.A., Brier, J.A., Chan, E., Chernov, J., Cohen, T., Deans, M., Garcia, A., German, C. R., Hauer, M., **Hu, S.K.**, Huber, J.A., Kane, R., Kobs Nawotniak, S., Lees, D., Lowe, J., Lubetkin, M., Marsh, L., Milesi, V., Miller, M., Miramalek, Z., Saunders, M., Sharif, K., Shields, A., Shock, E., Smith, A.R., and Sylva, S. (2020), SUBSEA 2019 Expedition to the Gorda Ridge, *Oceanography* 33(1), Supplement Pages 36 – 37.
8. Goordial, J., **Hu, S.**, & Tully, B. (2020). C-DEBI NextGen 2019 Early Career perspective on ‘What’s Next?’: Upcoming Challenges and Opportunities. DOI: 10.31219/osf.io/7xkqp

INVITED SEMINARS

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| 2026 | US Frontiers of Science Symposium, National Academy of Science, Irvine, CA |
| 2025 | College of Marine Science, University of South Florida, St. Petersburg, FL |
| | Aquatic Microbiomes on a Changing Planet, Barcelona, Spain |
| | Ocean Carbon & Biogeochemistry Workshop Summer Science Workshop, Woods Hole, MA |
| | Biology Department Seminar Series, Portland State University, Portland, OR |
| 2024 | American Society of Microbiology – Texas Branch Meeting, Galveston, TX |
| | School of Oceanography, University of Washington, Seattle, WA |
| | Nature Portfolio: Oceans Community, Ecology and Evolution meeting |
| 2023 | Horn Point Laboratory, University of Maryland Center for Environmental Science |
| | Department of Biology, University of Louisiana at Lafayette, Lafayette, LA |

- Marine Biology & Marine Science Seminar, Galveston, TX
- 2022 Marine Microbes Gordon Research Conference, Les Diablerets, Switzerland
- 2021 Department of Oceanography, Texas A&M University, College Station, TX
- BioGeoSCAPES scoping workshop
- OSU Microbiology Spring Research Symposium, Corvallis, OR
- Microbiology Society Annual Conference, UK
- University of Rhode Island GSO Department seminar, Narragansett, RI
- 2020 WHOI Biology Department, Woods Hole, MA
- WHOI Marine Chemistry & Geochemistry Department, Woods Hole, MA
- UGA Skidaway Department Seminar, Savannah, GA

CONFERENCE PROCEEDINGS (*denotes graduate student-led, #denotes undergraduate-led)

- 2026 #Sonsel, Meagan & **Hu, S.K.** *Interactive Data Portal for Microbes in the Northern Gulf*. GCOOS Spring Members Meeting, College Station, TX (Poster)
- #Day, A., Grimmer, D., Thornton, D., & **Hu, S.K.**, *Surviving Ocean Shifts: Diatom Response to Light Stress*. GCOOS Spring Members Meeting, College Station, TX (Poster)
- *Adams, A., Huber, J. A., **Hu, S. K.**, *A Little Vent Goes a Long Way: Protistan Community Composition in Hydrothermal Plumes*. Ecological Integration Symposium, College Station, TX (Oral)
- *Nedd, K., Huber, J.A., **Hu, S.K.** *Getting Hot in Here: Vent-Exclusive Protists at Hydrothermal Vents of the Axial Seamount and Mid-Cayman Rise*. Ecological Integration Symposium, College Station, TX (Oral)
- #Lerma, M. & **Hu, S.K.**, *The Ecology of Early Microbial Eukaryotic Settlers at Deep-Sea Hydrothermal Vents*. Ecological Integration Symposium, College Station, TX (Poster)
- *Woodward, H. & **Hu, S.K.** *Microbial Metazoa in the Northern Gulf and Their Prey*. Ecological Integration Symposium, College Station, TX (Poster)
- #Lerma, M. & **Hu, S.K.**, *The Ecology of Early Microbial Eukaryotic Settlers at Deep-sea Hydrothermal Vents*. Undergraduate Research Scholars Symposium, College Station, TX (Oral)
- 2025 #Lerma, M., Smith, A. R., Huber, J. A. & **Hu, S.K.** *Microbial Presence and Diversity on Microcolonizers at Deep-Sea Hydrothermal Vent*. Student Research Week, College Station, TX (Oral)
- #Seshampally, S., Campbell, L., Quigg, A., Henrichs, D., & **Hu, S.K.** *Investigating Abiotic Influences on Carbon Cell Abundances of Karenia brevis in the Gulf of Mexico*. Student Research Week, College Station, TX (Oral)
- #Sonsel, M., #Day, A., #Lerma, M., #Seshampally, S., *Nedd, K., *Adams, A., & **Hu, S.K.** *How do Environmental parameters impact protistan communities in the Gulf of Mexico*. Student Research Week, College Station, TX (Poster)
- #Day, A. & **Hu, S.K.** *Diatoms in the Deep: Can Photosynthetic Life Survive in the Deep ocean?* Student Research Week, College Station, TX (Oral)
- Hu, S.K.**, Smith, A., Anderson, R., German, C., Lang, S.Q., Butterfield, D., Fortunato, C.S., Seewald, J., & Huber, J.A. *Death in the Deep: Microeukaryotes in hydrothermal vent food webs*. Deep-Sea Biology Symposium, Hong Kong, China (Oral)
- 2024 *Adams, A., Huber, J. A., **Hu, S. K.** *Microbial eukaryotes throughout the plume at Axial Seamount*. Ecological Research Symposium (Poster)
- *Nedd, K., Huber, J.A., **Hu, S.K.** *Endemicity within deep-sea hydrothermal vent protists*. Ecological Research Symposium (Poster)
- Hu, S.K.**, Smith, A., Anderson, R., German, C., Lang, S.Q., Butterfield, D., Fortunato, C.S., Seewald, J., & Huber, J.A. *The elusive ecological roles of microeukaryotes at deep-sea hydrothermal vents*. Ocean Sciences Meeting, New Orleans, LA (Oral)
- #Day, A., Lerma, M., Seshampally, S., Sonsel, M., & **Hu, S.K.** *Database Darkly: Why do we care about protistan biodiversity in the deep sea?* Oceanography Department 75th Anniversary Symposium (Poster)

- 2022 Elkassas, S., **Hu, S.K.**, Anderson, R.E., Butterfield, D.A., Seewald, J., Huber, J.A., & Fortunato, C. S. *Comparing Active Subseafloor Chemolithoautotrophic Communities from Mafic and Ultramafic Deep-Sea Hydrothermal Vents as Models for Understanding the Potential for Life on Ocean Worlds*. Astrobiology Science Conference. Atlanta, Georgia (Oral)
- 2021 **Hu, S.K.**, Herrera, E., Smith, A., Seewald, J.S., German, C.R., & Huber, J.A. *Microbial eukaryotic diversity & grazing activity at deep-sea hydrothermal vents*. Deep Sea Biology Symposium. Brest, FR (Oral)
- 2020 #Herrera, E., Huber, J. Serres, M., **Hu, S.K.** *Protistan grazing in deep-sea hydrothermal vents*. Ocean Sciences Meeting, San Diego, CA (Oral)
- Chan, E. W., Breier, J. A., German, C. R., Huber, J. A., Kobs-Nawotniak, S. E., Shock, E., Raineault, N., Hauer, M., Krasnosky, K., Sylva, S., **Hu, S. K.**, Smith, A. R., Milesi, V. P., & Lim, D. S. S. *Going the Last Kilometer: Overcoming Challenges to Discovering Life-Supporting Gradients on Ocean Worlds*. Ocean Sciences Meeting. San Diego, CA (Oral)
- Hu, S.K.**, Herrera, E., Smith, A., Pachiadaki, M.G., Edgcomb, V.P., Sylva, S.P., Chan E.W., Seewald, J.S., German, C.R., & Huber, J.A. *Microbial eukaryotic diversity & grazing activity at deep-sea hydrothermal vents*. Ocean Sciences Meeting, San Diego, CA (Oral)
- 2019 #Herrera, E., Huber, J. Serres, M., **Hu, S.K.** *Protistan grazing in deep-sea hydrothermal vents*. Summer Student Fellow Program Symposium, Woods Hole, MA (Poster)
- Hu, S.K.**, #Herrera, E., Smith, A., Seyler, L., Trembath-Reichert, E., Serres, M. Sylva, S., Seewald, J., & Huber J. A. *Ecological roles of microbial eukaryotes at hot spots of primary production in the deep sea*. (2019) C-DEBI Annual Meeting. Monterrey, CA (Oral)

TEACHING

Biological Oceanography (OCNG 620) – TAMU

Professional Communication in Oceanography (OCNG 303) – TAMU

Scientific Writing in Oceanography (OCNG 689) – TAMU

The Blue Planet: Intro to Oceanography (OCNG 251) – TAMU

Introduction to R programming language, Workshop held in 2024, Woods Hole, MA

Advanced data visualization in R, Workshop held in 2021, Woods Hole, MA

Introduction to R programming language, Workshops held in 2019, 2020, 2021

Instructor for Bioinformatics Virtual Coordination Network, 2020-2021

PROFESSIONAL ACTIVITIES & SERVICE

2024 Metaeukomics – Co-lead for the Ocean Carbon Biogeochemistry small group activity

2021 URGE Leader & DEI actions at WHOI – Led discussion group

2019 C-DEBI Early Career Workshop – Organizer and participant for workshop

2019 NSF Ideas Lab EukHiTS – Workshop participant

2018 Ecological Dissertations in Aquatic Sciences – Participant, microbial group leader

2014-present Invited peer reviewer: *NSF BO, ISME Journal, Nature Communications, Molecular Ecology, Environmental Microbiology, Aquatic Microbial Ecology, mSystems, Deep Sea Research II, & Limnology & Oceanography*

RELEVANT OUTREACH

2026 Darwin Day – TAMU Ecology & Evolutionary Biology

2025 Expanding Young Horizons – LEVELUP Texas, TAMU

Youth Career Fair – TAMU

Texas Junior Academy of Science – TAMU

2024 Intermediate School Science Outreach – Jane Long Intermediate School